A Survey of Semantic Approaches in Personality Traits Profiling Analysis

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***Abstract*— Natural language processing (NLP) and machine learning have been used in the analysis of individual profiling based on various data sources, including public datasets, social media platforms, and frameworks such as DISC, Big Five, MBTI, Holland's RIASEC, 16 PF, HEXACO, and others. Deep learning is also utilized in research, with algorithms such as CNN, LSTM, and Galvanic Skin Response (GSR) employed for classification, labeling, and regression. The development of profiling analysis has led to a semantic approach to enhance model performance. Examples of new models that can improve performance include the Semantic-enhanced Personality Recognition Neural Network (SEPRNN) and the Robustly Optimized BERT Pre-Training Approach (RoBERTa). This study begins by collecting papers from publication databases. The next step is paper selection, and then the paper content is analyzed according to the research questions that have been determined. The final step is visualization. This study aims to provide a comprehensive understanding of the development of NLP in profiling analysis. The results of this study show that 53% used social media as a data source in profiling analysis, 43% used deep learning for the learning model, 68% used the Big-Five personality framework, 10.2% used support vector machine (SVM) as the algorithm, and 86.7% used lexical semantics for approaches. Research opportunities in profiling analysis include developing and improving predictive models, integrating and exploring NLP, using diverse data and machine learning techniques, applying them in various contexts, and integrating them with other technologies.**

Keywords—semantics, personality, traits, profiling, analysis

I. INTRODUCTION

Natural Language Processing (NLP) and Machine Learning, when applied in the field of psychology, have produced various new methods for analyzing personality profiling characteristics of individuals [1], [2]. The data used for profiling analysis comes from various sources, including public datasets like those from Kaggle, private repositories, and data from social media [3], [4]. Social media platforms such as twitter (X) [5], facebook [2], instagram [6], and linkedIn [7] have become popular data sources for developing models in profiling analysis. The type of data used in profiling analysis varies depending on the social media platform; for example, many studies on twitter (X) use text data for profiling analysis [8], while instagram uses image data [6]. In psychology, there are many personality frameworks used for personality analysis, including DISC [9], BigFive [10], MBTI [11], Holland's RIASEC [12], 16 PF [13], HEXACO [14], and others. Popular frameworks used in many studies include BigFive, Myer Briggs Type Indicator (MBTI), Dominance-Influence-Steadiness-Compliance (DISC), and HEXACO. The choice of research framework is tailored to each study's needs and objectives. For example, a study [15] used the DISC framework to analyze the personalities of job seekers based on their twitter account posts. In contrast, a study [16] used the MBTI framework for mapping employee personalities with job description requirements. Another study [17] used the BigFive framework to understand the relationship between semantic features in text and talent management opportunities.

Although many profiling analysis studies use machine learning, many also employ deep learning. For instance, a study [18] used deep learning with convolutional neural network (CNN) algorithms for personality prediction based on handwriting analysis. Similarly, a study [19] used deep learning with CNN and long short-term memory (LSTM) algorithms to predict social media users' personalities based on their text posts. Another study [20] used deep learning with the GSR algorithm for sentiment recognition and personality detection. As the field evolves, profiling analysis research moves towards a semantic approach to improve model performance. According to many studies, semantic approaches can enhance model performance in classification, clustering, and regression. For example, a study [21] proposed a new model called the semantic-enhanced personality recognition neural network (SEPRNN), which can improve performance in recognizing personality traits based on text. The performance of SEPRNN has surpassed previous models, especially in personality traits with multiple labels. Study [22] also proposed a new model called the robustly optimized BERT pre-training approach (RoBERTa). Combining RoBERTa, artificial neural network (ANN), and SMOTE with a semantic approach can improve the accuracy of personality predictions [23].

A Systematic Literature Review (SLR) aims to provide an overview of the development of the NLP field in the topic of Profiling Analysis with a semantic approach. This survey offers valuable insights into future research directions, model performance accuracy improvements, and industrial applications, such as recruitment, personalized marketing, and mental health assessment. Overall, this paper provides valuable insights into how increasingly advanced NLP and machine learning techniques can be used for personality profile analysis, with particular emphasis on the role of semantic approaches in optimizing the results of such analyses. It offers an in-depth look at the potential and future research directions in this area, underscoring the need for

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more innovative development models and the integration of more advanced NLP techniques.

II. Background Theory

A. Profiling Analysis

Profiling Analysis is a multidisciplinary field encompassing computer science, psychology, and data analysis to predict individual personality traits based on datasets [24]. The application of Profiling Analysis in social media refers to the process of collecting, analyzing, and interpreting data available on social media platforms to understand user behavior, preferences, and characteristics [9], [24], [25]. This often involves analyzing text, images, and social interactions, such as likes, shares, and comments [9]. One of the primary applications of profiling analysis in social media is marketing, where companies use the collected data to understand their customers' habits and preferences, targeting advertisements and content more effectively [24]. A study by Erevelles et al. [26] shows that data from social media can provide deep insights into consumer preferences and habits, which cannot be obtained through traditional methods. Beyond marketing, profiling analysis in social media is also used in social and psychological behavior research. This research often focuses on how identity and self-expression are formed and maintained through interactions on social media [27]. For example, a study by Marengo D et al. explored how facebook profiles influence users' self-esteem [28]. This research found that positive social media interactions can significantly impact an individual's self-image and emotional well-being. This fact demonstrates the importance of social media profile analysis not only for commercial purposes but also for understanding the psychosocial dynamics of individuals in the digital era. These two studies show the broad and diverse applications of profiling analysis in the context of social media, highlighting its significance in understanding modern human behavior.

B. Personality Traits

Personality traits are various characteristics that form an individual's personality, manifested in relatively consistent patterns of thought, feeling, and behavior that distinguish one individual from another. There are many frameworks for categorizing different types of personality traits. The most commonly used framework is the Big Five, which groups personality traits into five dimensions: openness, conscientiousness, extraversion, agreeableness, and neuroticism [29][30]. Other research uses the DISC framework to categorize personality traits into four types: Dominance, Influence, Steadiness, and Conscientiousness [9][25]. Besides the Big Five Framework and DISC, the MBTI framework divides personality into sixteen types, and HEXACO divides personality into six types [31][32]. Many researchers use machine learning methods to automate the detection of a person's personality based on social media posts, both from text and image data. For example, in research by Jeremy N et al. [33], LSTM, Bi-LSTM, and GRU algorithms are used to identify language usage trends and predict a person's personality traits. Its implementation can be found in many fields, such as the study conducted by Ren Z et al. [34], understanding personality traits associated with emotional stability and the quality of relationships formed in social medicine. Furthermore, more comprehensive implementations have been carried out in research conducted by [35] and [17]. Personality studies combine traditional psychological theory with modern computational methods, especially in NLP.

C. Semantic Approach

The semantic approach in profiling analysis offers a unique perspective in understanding and analyzing data, particularly in comprehending the meaning and context behind collected data. This approach is highly relevant in text analysis, where words or phrases' meaning, nuances, and context are analyzed to gain a deeper understanding of the subject [36]. For example, in research conducted by Ghadekar et al. [37], a semantic approach was used in developing one of the modules of an automatic interview system that utilizes NLP and deep learning techniques in the job recruitment process. This system includes three modules, one of which is the resume classification module, where the semantic approach helps calculate contextual semantic similarity between two strings of words or phrases, which is an essential part of the resume classification and interview question process. Furthermore, the semantic approach also plays a significant role in social media analysis, as discussed in research by [38]. This study developed a system for predicting Big Five personality traits in twitter users. The research combined SVM and BERT (Bidirectional Encoder Representations from Transformers) as semantic approaches. Additionally, linguistic inquiry word count (LIWC) was explored as a linguistic feature for personality prediction. The main goal of this study was to improve the performance of the personality prediction system by implementing these methods. Moreover, the semantic approach was also used in research [39] that developed and evaluated a spam detection model using deep learning to identify and filter spam on social media platforms, specifically twitter. The semantic approach was employed to extract features based on the topics of twitter posts, which helps detect spammers on twitter.

III. Methods

This research employs a systematic approach to review the literature on the development of the NLP field, specifically focusing on Profiling Analysis with a semantic approach. The review mainly centers on the data sources, learning models, personality frameworks, algorithms, and the semantic approaches utilized in this area of research.

A. Research questions (RQ)

RQ1 : How frequently are specific keywords in the literature related to our research area?
RQ2 : What are the primary data sources used in existing studies within our field?
RQ3 : Which learning models are most commonly employed in recent studies in our domain?
RQ4 : What are the significant personality frameworks used in existing studies within our field?
RQ5 : What algorithms are predominantly used in our research area?
RQ6 : In what ways are semantic approaches applied in the literature?

B. Study Selection

The research articles included in this Systematic Literature Review (SLR) were selected from 2020 to 2023, with additional papers from 2019 and 2024. We searched databases from publishers, including IEEE Xplore, Science Direct, Springer, Emerald, ACM Digital Library, Scopus, and SAI.

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The keywords used for searching papers include 'Profiling Analysis,' 'Personality Prediction,' and 'Semantic Approach Personality.' The collected papers were then selected based on their titles and abstracts. The first step is to choose a digital library that will be used to search for literature and then determine search keywords that are relevant to the specified research topic. Research papers or articles are downloaded from the digital library using predetermined keywords. The downloaded papers are then analyzed to understand the content and arguments presented. Next, analyze the research results from the paper to extract meaningful information. The analyzed papers were extracted and arranged in SLR tables. The final step is to create data visualization and draw conclusions. The overall workflow can be seen in Figure 1.

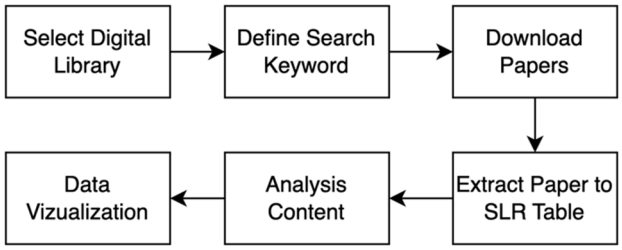


Fig. 1. Systematic Literature Review Study Flow

IV. Result and Discussion

A. Frequency Analysis of Keywords

Based on the problem formulation in RQ1, in the abstracts of the collected papers, the word 'personality' appears 95 times, prominently indicating that the main focus of the papers is closely related to the concept of personality. The high frequency of this word suggests a likely in-depth exploration of how personality can be studied, analyzed, or predicted.

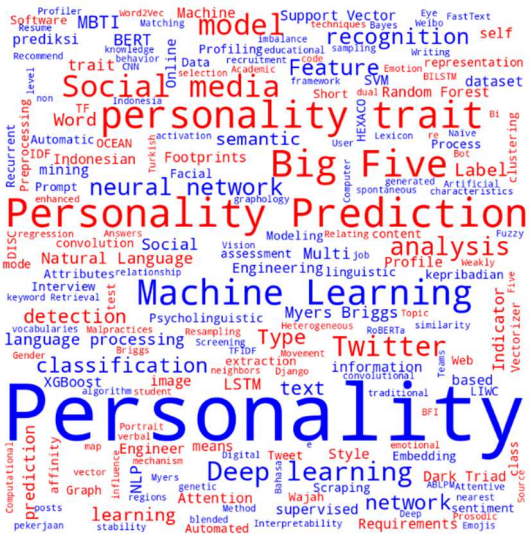


Fig. 2. Wordcloud frequency keywords

The words 'learning' and 'prediction' appear 37 and 30 times, respectively, indicating a focus on learning processes and predictions within the context of this research, likely related to machine learning or behavioral prediction. The words 'traits,' 'machine,' 'big,' and 'five,' appearing 25, 23, 22, and 23 times collectively indicate a connection between personality traits, machine learning technology, big data analysis, and the Big Five personality model. The social and media aspects, indicated by the frequency of the words 'social' (19 times) and 'media' (15 times), along with 'twitter' and 'analysis' appearing 13 and 11 times, respectively, point to the likelihood that the papers explore how personality is manifested or can be analyzed through social media, particularly twitter. The words 'deep' and 'model,' which also appear 13 times, affirm the possibility of using deep learning models in this analysis. Overall, the frequency of these words suggests that the paper's abstracts focus on integrating personality psychology, advanced technologies such as machine learning and big data analysis, and their applications in social media, particularly for understanding and predicting twitter user behavior, as illustrated in Figure 2.

B. Data Source

Based on the provided data, the most dominant data source is Social Media, with a significant proportion of 53.8%. This indicates that social media platforms are highly relied upon in data collection due to their broad reach and ease of access to user data. Online repositories also play an important role, contributing 19.2%, indicating that this source is valued for its potential to store structured and easily accessible data. Online Tests, contributing 15.4%, show that web-based assessment tools are also a reliable data source, possibly due to their ease of use and ability to collect data on a large scale. Personal Data, though only comprising 5.1%, remains essential, possibly containing more specific and sensitive information. Meanwhile, other data sources such as Community Platforms, Multimodal Databases, Online Platforms, Online Tests/Interviews, and Websites contribute only 1.3%, indicating that they are less prioritized or perhaps less accessible than the primary sources mentioned earlier. Based on the problem formulation in RQ2, this data indicates a clear preference for using digital platforms for data collection, particularly on social media and online repositories, as illustrated in Fig. 3.

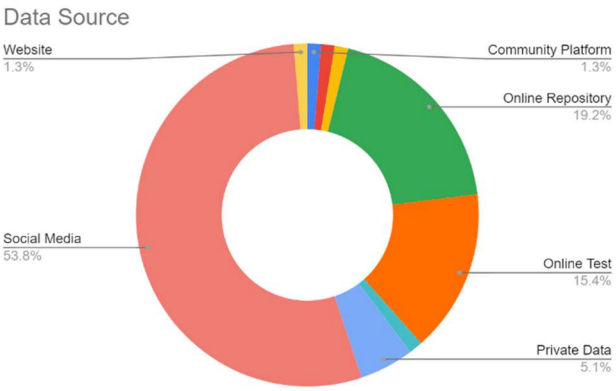


Fig. 3. Profiling analysis datasource

C. Learning Model

Deep learning is the most dominant learning model, accounting for almost half the total usage, at 49.4%. This reflects the current trend where deep learning is becoming increasingly popular, especially in applications that require complex data processing, such as speech recognition, natural language processing, and computer vision. However, traditional machine learning still has a strong presence, with a nearly equivalent percentage of 44.6%. This indicates that classic machine learning techniques remain relevant and

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widely used, particularly when more uncomplicated and interpretable models are desired. Lastly, the combination of machine learning and deep learning, though only 6%, suggests that there are specific cases where integrating both approaches yields better results, allowing the utilization of both methods to benefit from the best of both worlds. Overall, this discussion answers question RQ3, confirming that deep learning and machine learning are integral to the current technology landscape, with deep learning slightly more dominant, as illustrated in Fig. 4.

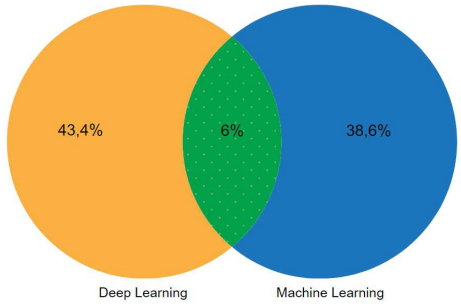


Fig. 4. Learning models

DL	:	[4], [18], [19], [20], [21], [23], [24], [29], [32], [34], [35], [37], [38], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66]
ML	:	[8], [9], [10], [11], [14], [16], [17], [25], [27], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93]
ML&DL	:	[3], [94], [95], [96], [97]

D. Framework Personality

Based on the problem formulation in RQ4, the most famous personality framework used is 'Big-Five,' which comprises 68.8% of the total. In comparison, other frameworks like 'MBTI' also show a significant portion with 15%, placing it second in popularity. 'Dark Triad Personality,' 'HEXACO,' and 'DISC' all have smaller shares, ranging between 2.5% to 3.8%, indicating that they are used but not as popular as 'Big-Five' or 'MBTI.' Lastly, 'Proactive Personality' has the smallest share in the chart, only 1.3%, indicating that it is the least used framework among those shown, as illustrated in Fig. 5.

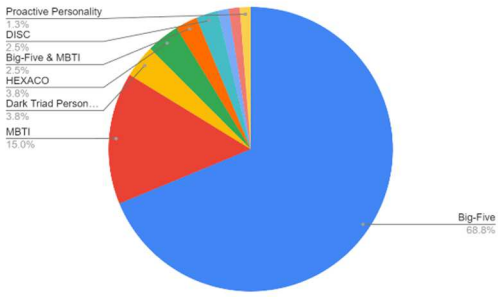


Fig. 5. Framework personality

Big-Five	:	[4], [10], [17], [20], [21], [23], [24], [29], [30], [32], [33], [35], [37], [38], [40], [41], [42], [43], [45], [46], [47], [48], [49], [51], [54], [55], [56], [57], [58], [60], [61], [63], [64], [65], [66], [67], [71], [74], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [88], [89], [91], [92], [93], [95], [96]
MBTI	:	[3], [11], [16], [19], [50], [59], [62], [69], [72], [87], [94], [97]
HEXACO	:	[14], [68], [73]
DISC	:	[9], [25]
DTP	:	[6], [8], [52]
PP	:	[90]
Big-Five& MBTI	:	[34], [53]

E. Algorithm

Based on the problem formulation in RQ5, SVM algorithm is the most frequently used in profiling analysis, with 10.1% of total applications. This is followed by the CNN and Random Forest (RF), each accounting for 8.9% of usage. Logistic Regression (LR) also holds a significant position with 7.6% usage. The LSTM and K-Nearest Neighbors (KNN) algorithms also hold important places with 6.4% and 5.7% usage, respectively, reflecting the importance of deep learning techniques in this field. Although only accounting for 4.4% of usage, BERT remains an important algorithm, demonstrating its effectiveness in specific tasks. XGBoost, Gradient Boosting (GB), and KMeans, with usage ranging from 1.9% to 2.5%, indicate that although they are not as popular as the main algorithms, they still play a role in profiling analysis. Meanwhile, algorithms with lower usage, such as AdaBoost, Decision Tree, and Multi-Layer Perceptron (MLP), each with around 1.3% usage, maybe more suited for specific cases or certain data conditions, as illustrated in Fig. 6.

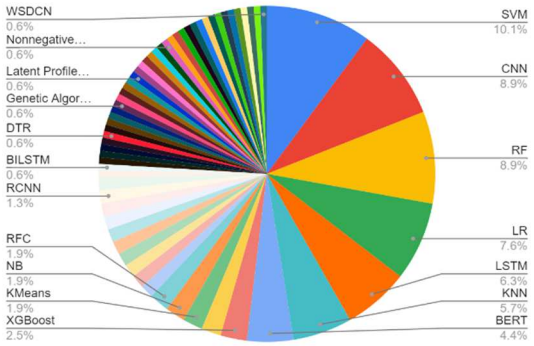


Fig. 6. Algorithm used for personality prediction

SVM	:	[8], [9], [10], [17], [25], [29], [38], [47], [59], [67], [74], [82], [85], [90], [96], [97]
CNN	:	[18], [19], [20], [33], [37], [40], [42], [43], [50], [51], [52], [53], [57], [81], [91], [96]
RF	:	[3], [29], [47], [67], [73], [74], [76], [79], [80], [83], [85], [88], [92], [96], [97]
LR	:	[29], [58], [59], [67], [77], [79], [80], [81], [82], [87], [96], [97]
LSTM	:	[4], [19], [20], [24], [32], [41], [52], [55], [56], [61], [62]
KNN	:	[3], [17], [37], [43], [47], [59], [81], [82], [96]
BERT	:	[32], [34], [35], [37], [38], [53], [60], [61]

F. Major Semantic Approach

Based on the problem formulation in RQ6, this research divides the semantic approach into lexical, cognitive, and structural [98]. Next, we analyze each paper to determine the semantic approach used in each paper based on the data and how it is processed. Lexical Semantics is the most dominant approach, occupying 81.5% of the total. This includes 75% purely in Lexical Semantics, 5.3% shared with Cognitive Semantics, and 2.6% overlapping with Structural Semantics, plus an additional 3.9% that might be a separate classification. Cognitive Semantics, although smaller, still plays a significant

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role with a total of 14.5%, divided between 9.2% exclusive and 5.3% overlapping with Lexical Semantics. Meanwhile, Structural Semantics accounts for 6.5% of the total, with 3.9% purely within it and 2.6% intersecting with Lexical Semantics. There is no interaction between Cognitive and Structural Semantics, and no part covers all three approaches simultaneously, indicating that although there is overlap, each approach has unique characteristics. Figure 7 illustrates the complexity and interactions in semantic studies in a very visual and informative way.

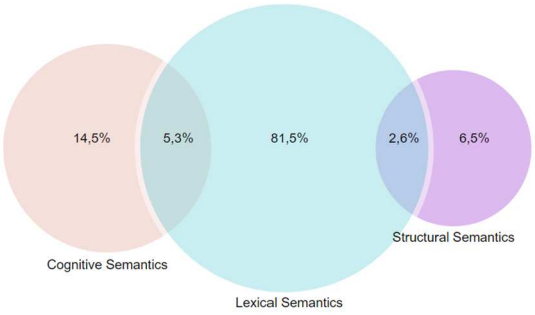


Fig. 7. Semantics approach

LS	:	[8], [9], [11], [14], [16], [17], [19], [20], [21], [23], [25], [27], [29], [32], [33], [34], [35], [37], [38], [40], [41], [42], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [59], [61], [62], [67], [68], [69], [70], [71], [72], [73], [74], [77], [78], [80], [81], [82], [83], [85], [87], [89], [90], [92], [96], [97]
CS	:	[6], [65], [66], [76], [79], [88], [91]
SS	:	[24], [43], [44]
LS&CS	:	[3], [60], [93], [95]
LS&SS	:	[64], [84]

V. Conclusion

Several key findings were obtained from the literature review conducted on profile analysis. First, Social Media emerges as the primary data source in profile analysis, dominating with a significant proportion of 53.2%. This indicates the crucial role of this platform in collecting data for user behavior analysis. Second, regarding methodology, deep learning is the most widely used learning model, accounting for half of the total usage. This signifies a strong preference for more advanced learning techniques in profile analysis. Third, the 'Big-Five' personality framework is the most popular, with its application covering 68.4% of the total. This reaffirms its status as a standard tool for understanding personality dimensions. Fourth, regarding specific algorithms, the SVM is the most frequently used algorithm, with a 10.2% percentage indicating its effectiveness in profile analysis. Lastly, Lexical Semantics is the dominant category, covering 86.7%, emphasizing the importance of language analysis in understanding and analyzing individual profiles. From the literature review results, several research opportunities emerge, including the development and improvement of predictive models, integration and exploration of NLP, use of diverse data and machine learning techniques, application in various contexts, and integration with other technologies.

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